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SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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CO5 - AT3 - HACKATHON

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QUESTIONS

Q1
CO5 - AT3 - HACKATHON / MINIPROJECT
100 marks

1/1 answered

Q1 **CO5 - AT3 - HACKATHON / MINIPROJECT** 100 marks

CO5 – AT2_HACKATHON

1. A satellite continuously sends telemetry packets containing temperature, battery level, fuel status, altitude, and communication status to a ground station. Thousands of packets may arrive simultaneously from different processing channels. The system must store incoming records dynamically because the number of packets is unknown in advance.

Develop a C-based Real-Time Satellite Telemetry Processing System that:

- Uses dynamic memory allocation for telemetry records.
- Uses structures and pointers to manage satellite data.
- Uses functions to modularize packet processing.
- Uses POSIX threads to process multiple telemetry streams concurrently.
- Uses mutex/semaphore synchronization to prevent simultaneous updates to shared telemetry data.
- Stores processed telemetry in a file for later analysis.
- Uses command-line arguments to specify satellite ID and input/output files.
- Uses IPC or socket programming to simulate communication between the satellite simulator and ground station.

Hackathon Challenge: The system must continue processing telemetry even when multiple packets arrive at almost the same time without losing or corrupting records.

2. A hospital emergency department receives patient requests from multiple registration counters simultaneously. Each patient record contains patient ID, emergency level, symptoms, assigned doctor, and treatment status.

Develop a Smart Emergency Patient Coordination System in C that:

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